

Junyi (Conny) Zhou

Boston, MA

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EDUCATION

- **Harvard University, T.H. Chan School of Public Health** Boston, MA
S.M. in Health Data Science *August 2025 – Expected December 2026*
 - Coursework: Deep Learning, Healthcare Machine Learning
- **Emory University, College of Arts and Sciences** Atlanta, GA
B.S. in Applied Mathematics and Statistics; Minor in Computer Informatics; GPA: 3.925 *August 2021 – May 2025*
 - Coursework: Data Structures, Database Systems, Machine Learning, Natural Language Processing

PROGRAMMING SKILLS

- **Languages:** Python, C++, Java, JavaScript, SQL, R
- **Frameworks:** PyTorch, Flask, React, scikit-learn, Hugging Face Transformers
- **Cloud & Data:** AWS (Lambda, EC2, S3, SageMaker), MongoDB, PostgreSQL, MySQL, Spark
- **Tools:** Git, Jupyter, high-performance computing
- **Certifications:** AWS Certified Cloud Practitioner, AWS Certified AI Practitioner, AWS Machine Learning Associate

TECHNICAL EXPERIENCE

- **Airway Management Simulation Platform** Atlanta, GA
Full-Stack Engineer, Scrum Master — Emory Pediatric Hospital *January 2024 – August 2025*
 - Built and deployed a clinical simulation platform using **Python/Flask, React, MongoDB**, and **AWS**, with resident-performance tracking and real-time analytics.
 - Integrated an **OpenAI RAG pipeline** with context management and hallucination controls to generate multi-turn training scenarios while protecting simulation data.
 - Led an Agile team of 6, conducting code reviews and incorporating clinician feedback to iteratively improve the application and NLP pipeline.
- **Transformer Architecture Reimplementation** Atlanta, GA
Machine Learning Researcher — Emory Dept. of Computer Science *October 2024 – January 2025*
 - Implemented a Transformer from scratch in **PyTorch**, including multi-head self-attention, positional encoding, layer normalization, and the training pipeline.
 - Tuned and benchmarked the implementation against `nn.Transformer`, validating correctness and achieving a **9.38 BLEU** score on English-to-German translation.
- **AlphaFold Protein–Nucleic Acid Interaction Pipeline** Atlanta, GA
Data Analyst — Bou-Nader Lab, Emory Dept. of Biochemistry *February 2025 – August 2025*
 - Automated high-throughput **AlphaFold** inference on HPC clusters, using parallel scheduling to process structural predictions across GPU nodes.
 - Developed **Python** scripts and CLI tools to filter and rank structures using pLDDT and PAE scores for wet-lab review, with documentation for non-ML researchers.
- **Cross-Species Drug Translation using Optimal Transport & Generative AI** Boston, MA
Graduate Researcher — Mooney Lab & Alvarez-Melis Lab, Wyss Institute *February 2026 – Present*
 - Engineered a custom **PyTorch CellOT** implementation using optimal transport and variational autoencoders to model cross-species treatment effects.
 - Designed **scanpy** data pipelines for automated cell-type labeling and integration of public and lab-generated single-cell datasets.